Middleware Assignment 1

We have the file and directory structure shown in the figure:

```
project/
│
├── data/
│   ├── sample_01/
│   │   ├── ranges_1.txt
│   │   ├── ranges_2.txt
│   │
│   ├── sample_02/
│   │   ├── ranges_1.txt
│   │   ├── ...
│   │   ├── ranges_n.txt
│   └── ...
│
├── results/
│
├── logs/
│
├── scripts/
│   ├── count_tot_primes.py
│   ├── ...
│
└── workflow_orchestrator.py
```

There is an indeterminate number of *sample* folders, and each of them contains several *range_x.txt* files. The *ranges_x.txt* files contain an indefinite set of numeric values of the type:

ranges_x.txt

```
 5    30
40   100
100  1000
```

The *count_tot_primes.py* script counts how many prime numbers there are in a range of two integers. The program's output consists of a list of the prime numbers found and their total.

```
$ python count_tot_primes.py  1 10
2
3
5
7
TOTAL:  4
```

From these elements, we want to obtain a single file (*ALL_PRIMES.txt*) containing all the prime numbers (without repetitions) generated from all the intervals contained in the *range_x.txt* files. The last line of the file will contain the total number of primes. Intermediate files and *ALL_PRIMES.txt* should be stored in the *results* folder.

We assume that running the *count_tot_primes.py* program is computationally intensive and therefore its execution should be done on the nodes of a cluster managed with SLURM.

You must write an *workflow_orchestrator* script (in Python) to efficiently solve this problem. The orchestrator's objective is to generate and control all the SLURM jobs that are needed to solve the problem. If needed, you can create other scripts to perform auxiliary operations (filtering duplicate values, sorting values, etc.). All these scripts will be located in the *scripts* folder. SLURM scripts must also be stored in the *scripts* folder. The *stdout* and *stderr* files automatically generated by SLURM must be stored in the *logs* folder. The orchestrator program must generate a log text file (*ORCHEST_LOG.txt*) showing the progress of the problem's solution (e.g., sample folders visited, ranges files used, SLURM jobs generated, etc.). This file should be also written in the *logs* folder.

Your solution will be evaluated based on the use of mechanisms that efficiently leverage available computing resources (e.g., xargs or parallel Linux commands, SLURM arrays, SLURM dependencies, parallel execution in SLURM, etc.).

The orchestrator program can be run in two ways:

$python workflow_orchestrator.py (run from within the *project* directory).
$python workflow_orchestrator.py path_to_project_folder (run from outside the *project* directory).

You must submit a PDF report describing the implemented orchestrator program and the auxiliary scripts used. The report should also describe the validation tests performed and, if applicable, any limitations or unresolved issues in the submitted version of the orchestrator. Along with the report, you must submit a compressed directory file with the structure shown in the previous figure (include several *sample* folders with *range_x.txt* files).